

Abstract — In an era of rapid digital onboarding and automated identity verification (e-KYC), traditional Optical Character Recognition (OCR) tools extract text but fail to verify physical or digital authenticity. **DocScreen** is an explainable, multi-layer artificial intelligence framework designed to detect identity document tampering, digital forgery, and template anomalies in real time. Combining computer vision forensics, layout consistency analysis, unsupervised anomaly detection (Isolation Forest), and cryptographic QR cross-validation, DocScreen fuses eight forensic signals into a deterministic **0–100 Risk Index** with transparent audit trails and human-in-the-loop decision governance.

1. The Challenge

Digital onboarding workflows are increasingly vulnerable to modified identity cards (Aadhaar, PAN, Voter IDs). Forgeries commonly exploit pixel-splicing, font mismatches, copy-move artifacts, and altered dates. Conventional verification pipelines rely strictly on static OCR or third-party database pings, missing localized visual modifications, invalid checksums, and synthetic document generators.

2. Proposed Solution

DocScreen implements a **defense-in-depth inspection pipeline**. Rather than relying on a single black-box classification, it subjects every document to an automated 13-step inspection sequence spanning visual noise analysis, coordinate-registered OCR, temporal validity rules, and ML-based anomaly models before issuing an explainable recommendation (Approved, Escalated, or Rejected).

3. MULTI-SIGNAL RISK FUSION ENGINE

Forensic Vector	Weight	Analytical Mechanism & Verification Technique
Visual Tamper & ELA	25%	Error Level Analysis (ELA), edge discontinuity, noise gradient variance, and copy-move detection.
Template & Layout Match	15%	Computer vision aspect ratio check, structural landmark alignment, and emblem/header validation.
Cross-Field Consistency	15%	Cross-checks DOB vs. Age, PAN card structure checksums, and inter-field semantic coherence.
ML Anomaly Detection	15%	Trained Isolation Forest evaluating multi-dimensional forensic features for outlier detection.
Dual OCR & Text Anomalies	10%	RapidOCR/Tesseract bounding-box mapping, character confidence scores, and font size consistency.
Pattern & Temporal Rules	10%	Regex pattern syntax, future issue dates, illogical expiry timestamps, and sequence verification.
Metadata & EXIF Forensics	5%	Detects image editing software tags (Photoshop, GIMP), software signatures, and missing EXIF profiles.
Image Quality Screening	5%	Laplacian blur variance, contrast balance, and glare/specular reflection analysis.

4. KEY INNOVATIONS & IMPACT

- Explainable AI (XAI):** Provides investigators with granular bounding boxes, risk signal attribution, and confidence scores rather than opaque classifications.
 - Dual OCR Redundancy:** Employs primary neural OCR with fallback text extraction engines for robust recognition in low-resolution scans.
 - Offline-First Scalability:** Functions entirely without external third-party credential databases, safeguarding citizen privacy and data sovereignty.
- Human-in-the-Loop (HITL):** Automated Risk Queue escalates borderline documents (31–65 Medium Risk) for secondary reviewer adjudication.
 - High Throughput:** Asynchronous microservice architecture delivers full multi-engine forensic evaluation in sub-3 seconds per scan.
 - Production-Ready Stack:** FastAPI (Python 3.11) backend with containerized OpenCV/Tesseract, Next.js 16 UI, and cloud-native deployment.